PRODUCT PERFORMACE UPDATE

Application: Metex, Ontario Canada – Allcase Furnace

ALLCASE FURNACE DETAILS

- **Furnace:** Allcase radiant tube, oil quench heat treat furnace operating at 1,600^0 F, used to heat treat various parts.

- **Lining Configuration:** The current ling consists of 2,000^0 F IFB on the sidewall and arch roof

- **Customer Need:** ITC recommended a solution to address several major challenges,
  - High fuel consumption.
  - Continual radiant tube weld repair.
  - Continual refractory repair and maintenance.

ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended a hot face coating to address the customers' requirements. The coating design included a proprietary high temperature coating, ITC 100HT, applied to the hot face of the IFB and ITC 213 metal coating applied to the radiant tubes.

- **High Temperature, Energy Efficient Ceramic Coatings:** ITC manufactures several proprietary, high temperature, energy efficient ceramic coatings to work in concert with refractory or metal substrates. Under ideal conditions, high these coatings absorb energy from the process and re-radiate it to the load, imparting the following benefits;

  Lower heat loss – Minimizes heat transfer through the refractory lining
  Lower Maintenance costs – Protects the substrate refractory
  Reduced preheating time – Improves the thermal efficiency of the process thus allowing quicker furnace start-up after downturns.

IMPACT ON THE PROCESS

- REFRACTORY LONGEVITY
  With the use of ITC Coatings refractory life is extended on average 2 to 3 times longer

- REDUCED PREHEAT TIME
  Start-up time to operating temperature was reduced by 50%

- IMPROVED FUEL SAVINGS
  Due to the thermal efficiencies, fuel consumption was reduced by 30%

- REDUCED RADIANT TUBE MAINTENANCE AND EXTENDED LIFE
  With the use of ITC Metal Coating, radiant tube life was extended by 2x